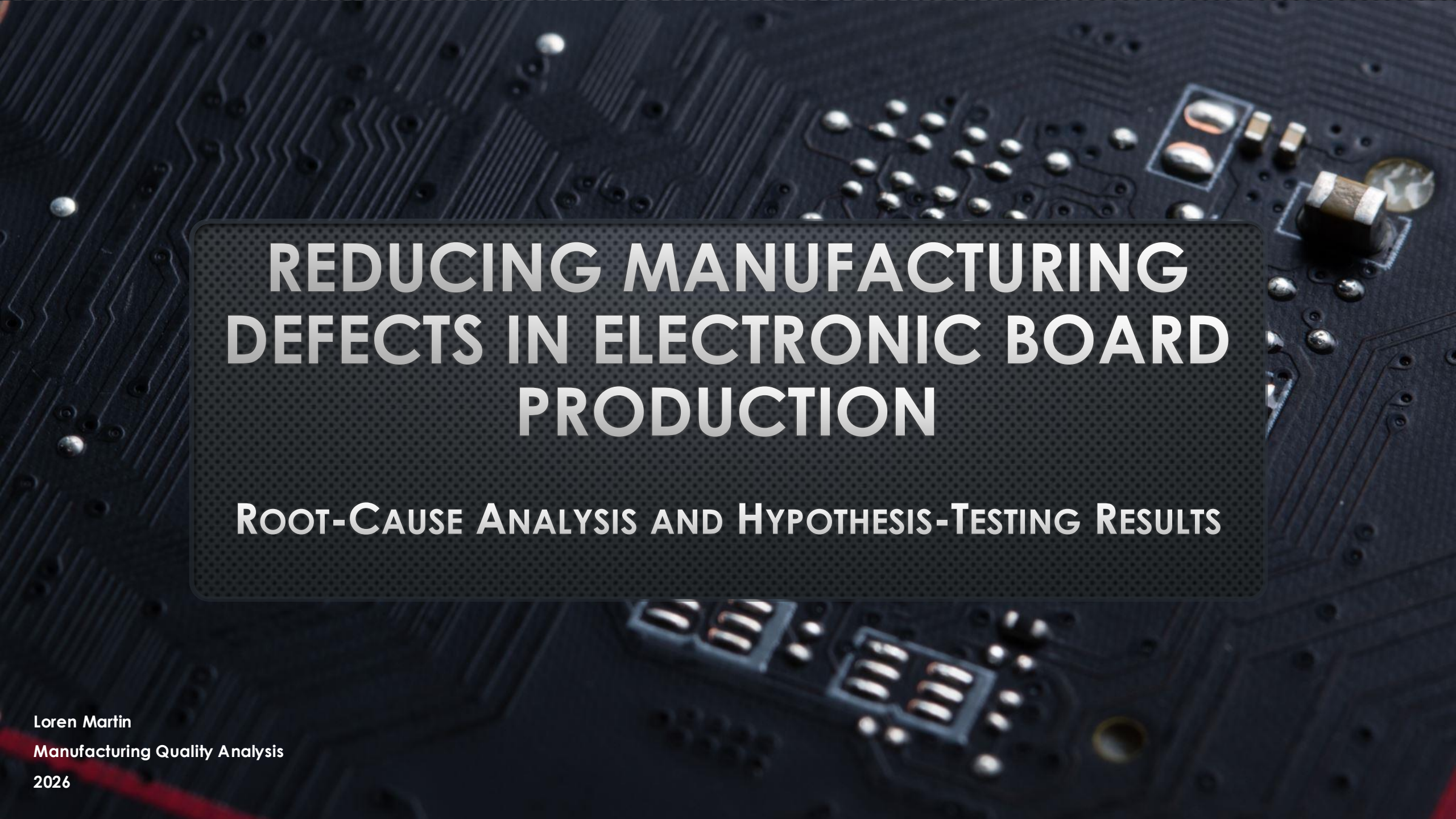




REDUCING MANUFACTURING DEFECTS IN ELECTRONIC BOARD PRODUCTION

ROOT-CAUSE ANALYSIS AND HYPOTHESIS-TESTING RESULTS

Loren Martin

Manufacturing Quality Analysis

2026

RISING DEFECTS THREATEN QUALITY, COST, AND PRODUCTION CAPACITY

CURRENT STATE



Welding and Thru-Hole defects are increasing while production demand is rising.



Defects create assembly and electronic test problems, costly rework, and additional labor requirements.



The analysis examined defect frequencies by production model and defect category in the Manual Finish area.

TARGET STATE

20%

REDUCTION IN WELDING DEFECTS

20%

INCREASE IN PRODUCTION CAPACITY

WITHOUT INCREASING THE DEFECT PERCENTAGE

THREE METHODS ANSWERED DIFFERENT DECISION QUESTIONS

Pareto Analysis



Where are defects concentrated?

Fishbone Analysis



What potential process factors may contribute to the defects?

ANOVA

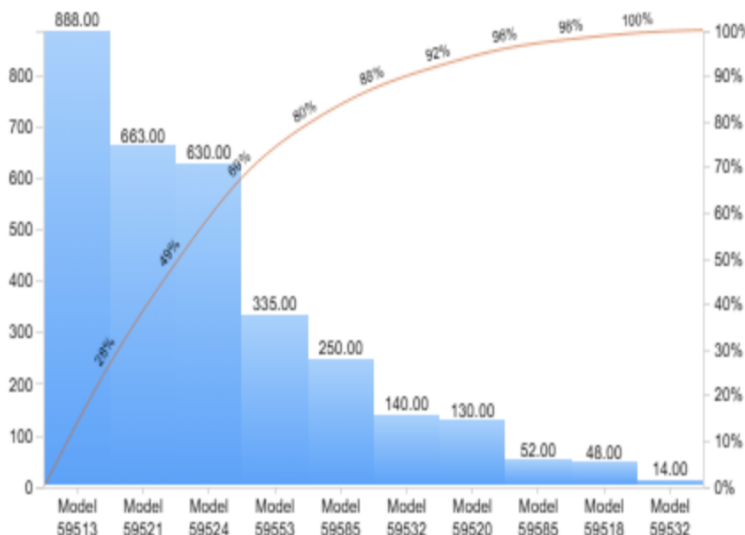


Do mean defect frequencies differ significantly among the three highest-defect models?

POWER BI DASHBOARD: KEY FINDINGS

Manufacturing Defect Analysis Dashboard

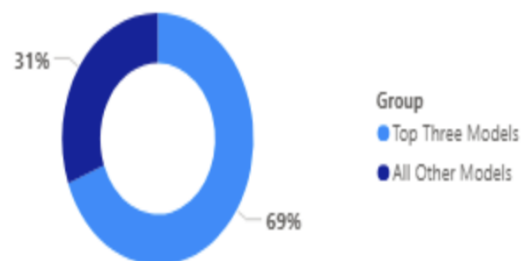
Defect Frequency by Production Model



Fishbone Diagram: Causes of Excessive Manufacturing Defects



Top Three Models Accounted for 69% of Defects

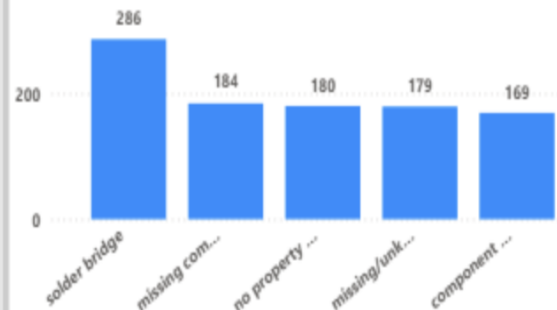


ANOVA Results

F-statistic = 0.579
Significance level = 0.05
p-value = 0.565
Fail to reject the H_0

No significant difference in mean defect frequency.

Most Frequent Defects in Top Three Models



Manufacturing Defect Dashboard

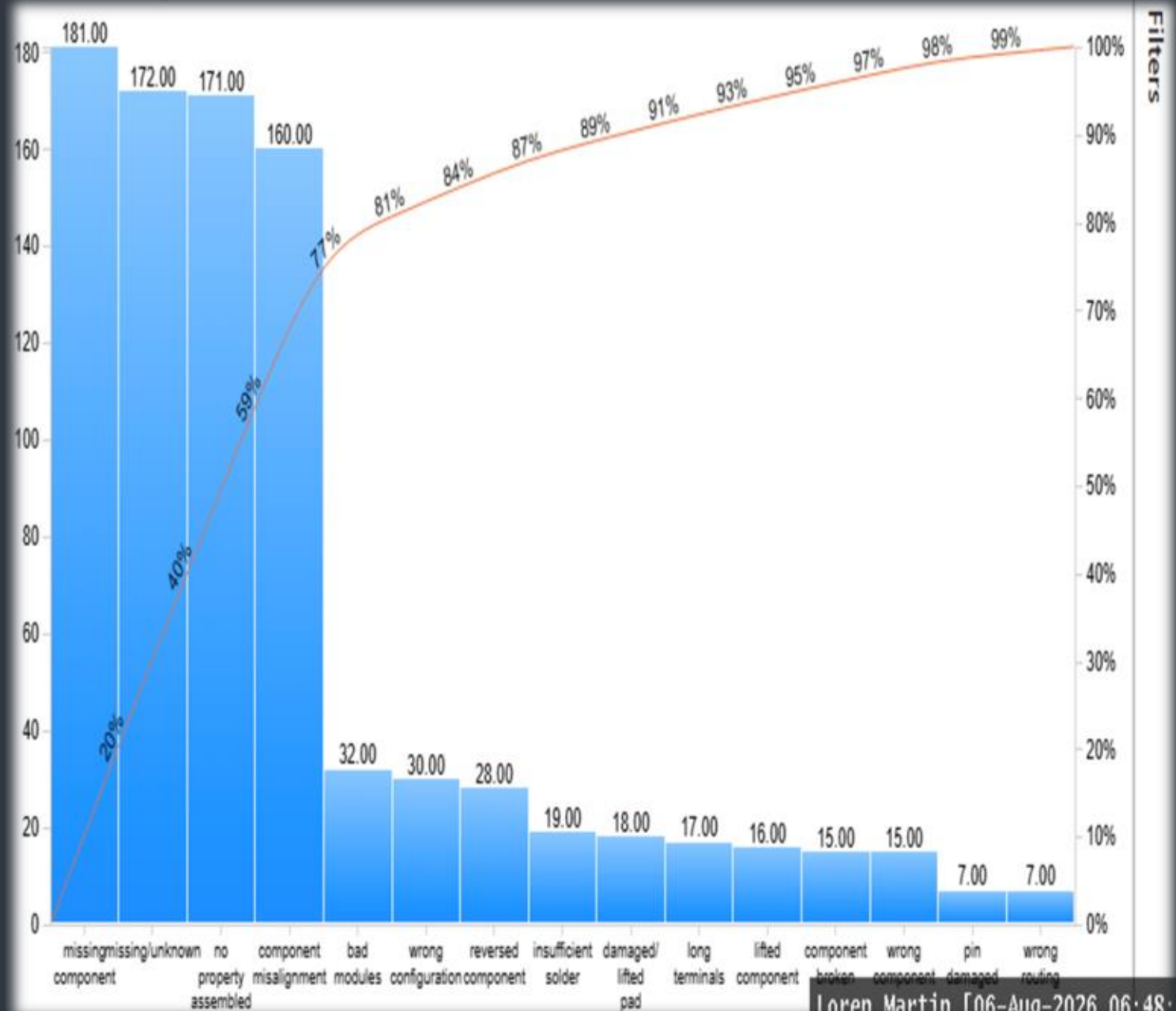
Model-Level Pareto Detail



Loren Martin [14-Aug-2026
Sandbox: NotebookBI

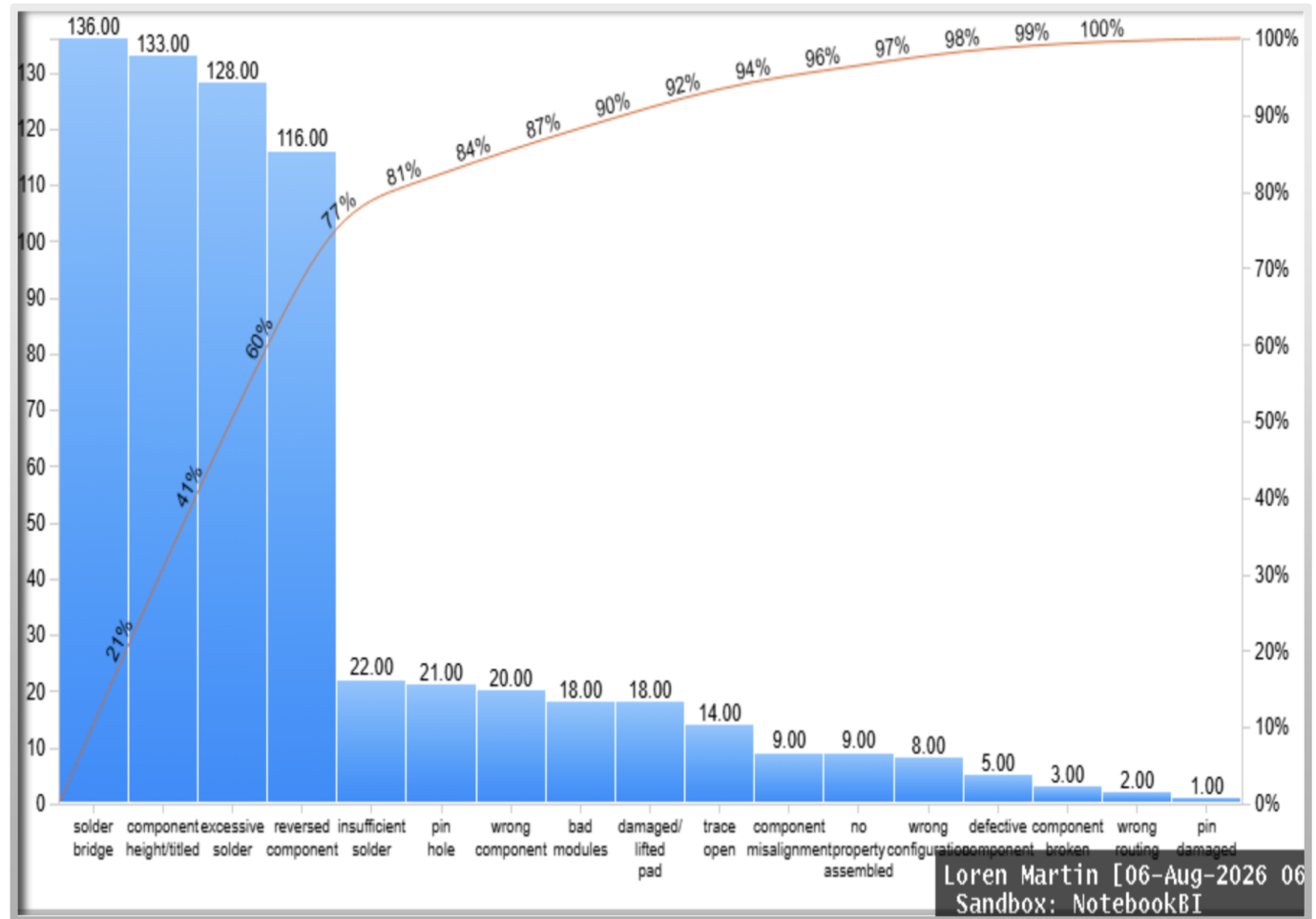
MODEL 595130 DEFECTS CENTER ON ASSEMBLY & COMPONENT PLACEMENT

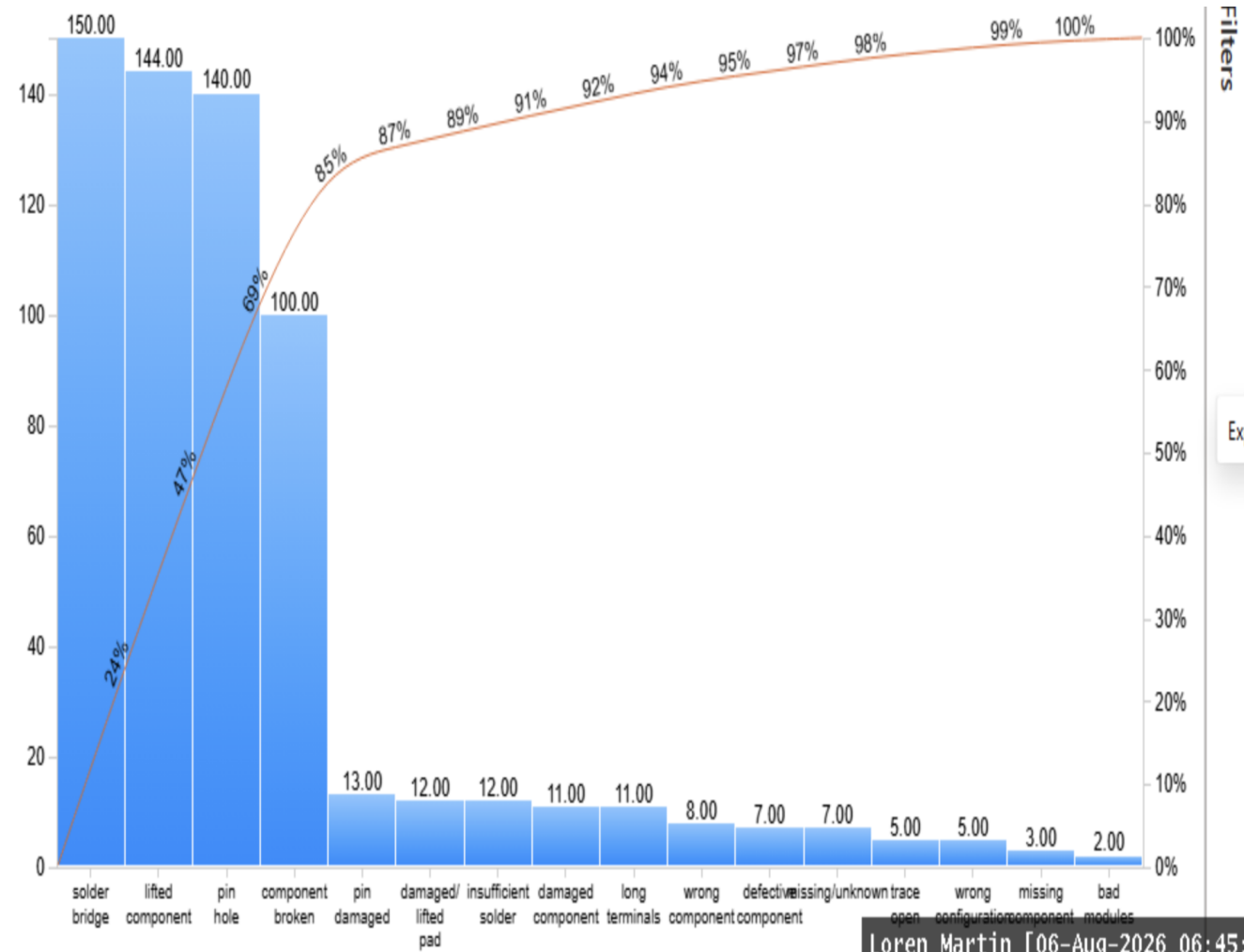
77%
OF MODEL 595130
DEFECTS CAME FROM
ITS FOUR LEADING
CATEGORIES.



MODEL 595214 SHOWS SOLDERING-RELATED DEFECTS

77%
OF MODEL 595214
DEFECTS CAME FROM ITS
FOUR LEADING
CATEGORIES.



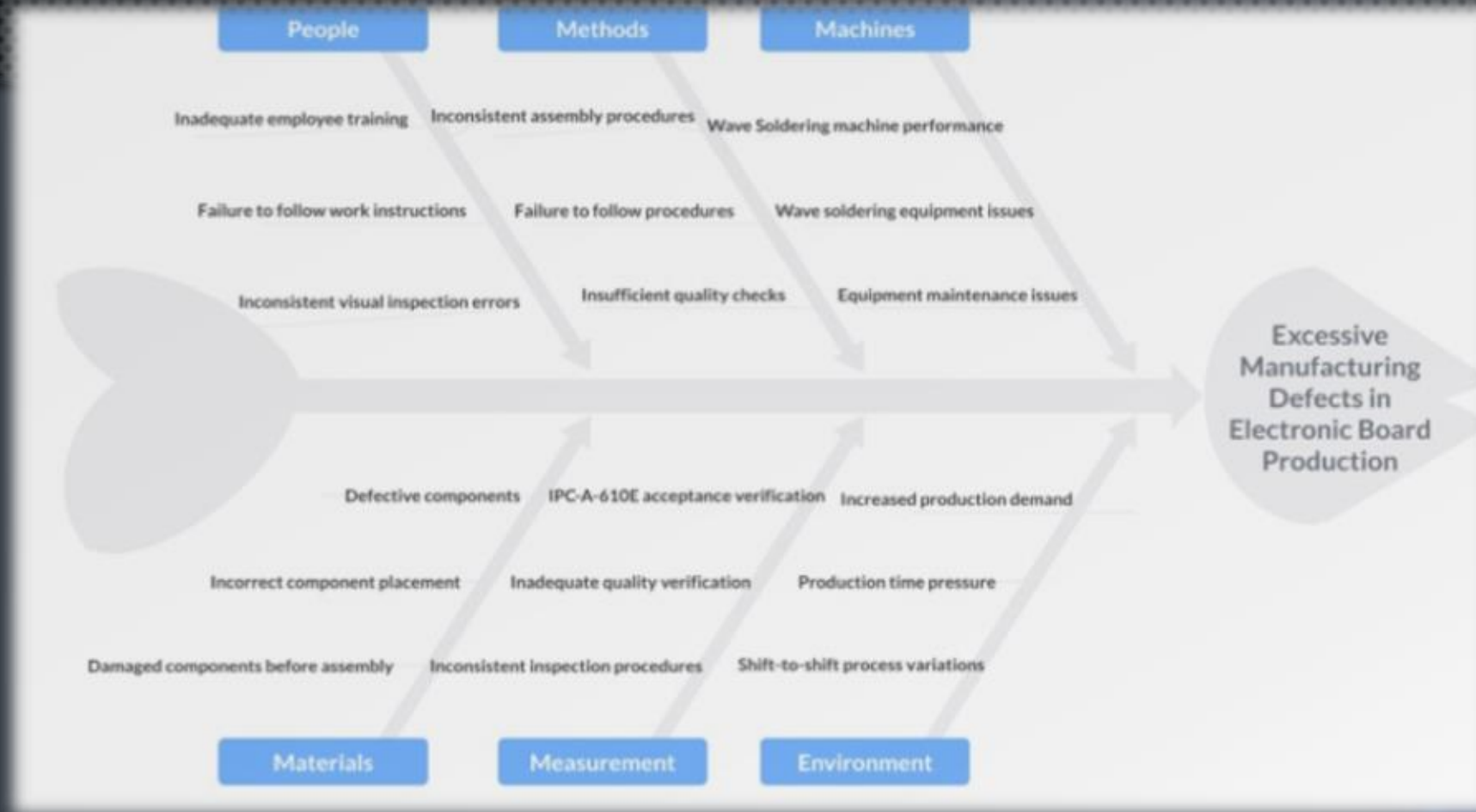


Loren Martin [06-Aug-2026 06:45:
Sandbox: NotebookRT

MODEL 595242
SHOWS
SOLDERING-RELATED
DEFECTS

85%
OF MODEL 595242
DEFECTS CAME FROM ITS
FOUR LEADING
CATEGORIES.

FISHBONE ANALYSIS HIGHLIGHTS POTENTIAL SHARED PROCESS RISKS



ANOVA FOUND **NO SIGNIFICANT DIFFERENCE** AMONG MODEL MEANS

```
from scipy.stats import f_oneway  
f_statistic, p_value = f_oneway(Model1_array, Model2_array, Model3_array)  
print(f"F-statistic: {f_statistic}")  
print(f"P-value: {p_value}")
```

F-statistic: 0.5789149585626375

P-value: 0.5646254542821236

Loren Martin [15-Jul-2026 03:42]
Sandbox: Notebook8I

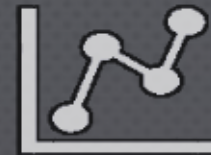


F-statistic: 0.579

p-value: .565

Significance level: .05

Decision: Fail to reject H_0



The analysis did not find sufficient evidence that mean defect frequency differed among the three production models.

COMBINED EVIDENCE SUPPORTS A PROCESS-WIDE RESPONSE



Pareto Analysis:
69%



Fishbone Analysis: 6
risk areas



ANOVA: $p = .565$

Decision Implication

- Investigate shared production conditions
- Prioritize high-frequency defects.
- Monitor each model with added attention to Model 595130.

PRIORITIZE FOUR PROCESS-WIDE IMPROVEMENTS

DESIRED OUTCOMES

20% Reduction in Welding Defects

20% Increase in Production Capacity

Without Increasing the Defect Percentage

IMPROVEMENTS



Standardize Work Instructions



Verify Equipment Controls



Targeted Training and Inspection



Monitor Defects by Model and Category

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